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Reg No.:_____ Name:_____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Eighth Semester B.Tech Degree (R,S) Examination May 2024 (2019 Scheme)

Course Code: ECT468

Course Name: RENEWABLE ENERGY SYSTEMS

Max. Marks: 100 **Duration: 3 Hours**

PART A

Answer all questions, each carries 3 marks. Marks

1	List out the merits and demerits of biomass as a renewable energy source.	(3)
2	What are the limitations of using conventional energy sources.	(3)
3	Draw the structure of a commercial solar cell.	(3)
4	Compare grid interfacing with isolation and without isolation.	(3)
5	What are the topographical features required while installing wind power plants.	(3)
6	Draw the block diagram of the permanent magnet synchronous generator.	(3)
7	Write short notes on grid integration.	(3)
8	Explain the concept of islanding in power systems.	(3)
9	Name any three emerging technologies of the smart grid.	(3)
10	What are the different applications of SCADA systems.	(3)

PART B

Answer any one full question from each module, each carries 14 marks.

Module I

11	a)	Discuss the world energy scenario of using various sources of energy.	(7)
	b)	Explain how geothermal energy acts as a renewable source of energy.	(7)

OR

12	a)	Explain the different energy sources available in India for electricity generation.	(7)
	b)	Discuss the various applications of solar energy systems.	(7)

Module II

13	a)	Explain the working of Tandem Solar cell with a neat diagram.	(7)
	b)	With a neat block diagram, explain the MPPT technique.	(7)

OR

14	a)	Explain any two methods used for thin film deposition.	(7)

b) Explain PV inverters with DC to DC converters on the low-frequency side with isolation with a block diagram. (7)

Module III

15 a) Explain the different components of a wind turbine with a neat diagram. (7)

b) Write short notes on (i) SCIG systems, (ii). DFIG systems. (7)

OR

16 a) Explain the working of WECS with a block diagram. (7)

b) Discuss the features of TARP and WARP systems. (7)

Module IV

17 a) Explain any two energy conversion schemes used in renewable energy generation. (7)

b) Explain the issues in the integration of converter-based sources (7)

OR

18 a) Explain the need for island detection systems. (7)

b) Discuss the importance of network voltage management in power systems. (7)

Module V

19 a) Discuss the protocols used in smart metering. (7)

b) Explain the architecture of SCADA systems. (7)

OR

20 a) Explain the block diagram of the smart meter. (7)

b) Explain distribution management system. (7)

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